GPU GROMACS Howto Recipes Install Recipe MPI

Installing and Using GROMACS

GROMACS is a free and open-source software suite for high-performance molecular dynamics and output analysis.

You can learn about GROMACS here: https://www.gromacs.org/.

Install GROMACS with MPI

Select a version on the GROMACS website, then download and extract the tar ball.

```
cd ~
mkdir gromacs
cd gromacs
wget --no-check-certificate http://ftp.gromacs.org/pub/gromacs/gromacs-
2019.6.tar.gz
tar xvfz gromacs-2019.6.tar.gz
```

Load MPI and Cmake modules,

```
module load engaging/openmpi/2.0.3 cmake/3.17.3
```

Create build and install directories,

```
mkdir -p 2019.6/build
mkdir 2019.6/install
cd 2019.6/build
```

Use `cmake` to configure compiling options,

```
cmake ~/gromacs/gromacs-2019.6 -DGMX_MPI=ON -
DCMAKE_INSTALL_PREFIX=~/gromacs/2019.6/install
```

Compile, check and install,

```
make
make check
make install
```

Set up usage environment,

```
source ~/gromacs/2019.6/install/bin/GMXRC
```

Run GROMACS

Firstly, prepare for an input file. Refer to file formats. Here shows an example with an input file named `benchPEP-h.tpr` downloaded from this page. In these examples we have saved the input files in the `~/gromacs/bench/` directory.

Secondly, create a batch job script, for example, named `job.sh`.

This job script requests 2 nodes with a total of 8 CPU cores and 50GB of memory.

job.sh

```
#!/bin/bash
#SBATCH --job-name="production run"
#SBATCH --partition=sched_mit_hill
#SBATCH --constraint=centos7
#SBATCH --mem=50G
#SBATCH -N 2
#SBATCH --ntasks 8
#SBATCH --time=12:00:00


module purge
module load gromacs/2018.4

gmx_mpi=/home/software/gromacs/2018.4/bin/gmx_mpi

if [ -n "$SLURM_CPUS_PER_TASK" ]; then
    ntomp="$SLURM_CPUS_PER_TASK"
else
    ntomp="1"
fi


# setting OMP_NUM_THREADS to the value used for ntomp to avoid complaints from
gromacs
export OMP_NUM_THREADS=$ntomp
```

```
mpirun -np $SLURM_NTASKS $gmx_mpi mdrun -ntomp $ntomp -deffnm
~/gromacs/bench/benchPEP-h -v
```

Finally, submit the job,

```
sbatch job.sh
```

Refer to GROMACS user guide for more info.

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